A Translation Floor for Cross-Lingual Archives: Spelling and Content Analysis in *Imvo Zabantsundu*

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Abstract

In its 1884–1894 issues, *Imvo Zabantsundu* published a large section in isiXhosa and a smaller one in English. This study asks whether a multilingual encoder (LaBSE) can detect this difference once the language signal is removed from its embeddings, and whether modernising the spelling changes this. The newspaper's own translations convey the same content in both languages, so a classifier's accuracy on them, the floor, shows how well it tells the sections apart without much difference in content. Performance above this floor shows that the sections differ in content, mainly in subject matter. Modernising the spelling helps the model read, bringing translations closer, but does not detectably change the content difference between the sections: spelling affects how well the model reads the newspaper, not what it finds.

Keywords: historical text normalisation; multilingual sentence embeddings; isiXhosa

Introduction

Imvo Zabantsundu, founded by John Tengo Jabavu in 1884, was the first African-owned newspaper in South Africa. It published mainly in isiXhosa, with some English translations and other items (Schoots, Ngwendu, et al., 2023). NLP tools are mostly developed from modern text in high-resource languages (Joshi, Santy, et al., 2020), which gives researchers a reason to modernise the spelling of isiXhosa texts, for example changing *Tixo* and *umteto* to *Thixo* and *umthetho*. Historical normalisation is judged by spelling accuracy or task scores (Bollmann, 2019), rarely by whether it changes a conclusion, making it a "forking path" (Gelman & Loken, 2013).

Research question. *After removing the language signal from a multilingual encoder's embeddings, can a classifier still distinguish between Imvo's sections based on their content, and does modernising the spelling change the size of this difference?* **H1:** A classifier predicting which section a chunk (a 60-word piece of a page) comes from should achieve a higher balanced accuracy (BA, its mean accuracy over the two sections; chance is 0.5) on all test chunks than on translated pairs, which are chunks matched to their translation in the other section. Each pair shares the same content, so its BA is the floor: how well the classifier tells the sections apart without any difference in content. The difference between the two, named Δ, measures the content difference. H1 is supported if the 95% confidence interval (CI) of Δ, resampled over test issues, lies above zero. **H2:** Modernising the spelling should lower the floor, since the encoder reads modern text better, while leaving Δ unchanged within ± 0.02 (measurement invariance). Beyond this range, modernising changes the conclusion.

Background Historical isiXhosa. The archive's machine-read (OCR) text was corrected by native isiXhosa speakers and deliberately not spell-checked, as 1880s spelling differs from today's (Schoots, Ngwendu, et al., 2023). Spelling is not neutral: written isiXhosa was first shaped by missionary grammars made for learners rather than speakers (Maseko, 2017), and low-resourcedness is also a question of who builds tools and for whom (Nekoto, Marivate, et al., 2020). **Removing language.** LaBSE (Feng, Yang, et al., 2022) was trained to match translations in 109 languages, isiXhosa among them, but its embeddings still identify the language. This signal can be removed by subtracting each language's average embedding (mean-centring; Libovický, Rosa, et al., 2020), which also removes average content differences, or by projecting out a few directions (Yang, Yang, et al., 2021; Ravfogel, Elazar, et al., 2020). As with control tasks (Hewitt & Liang, 2019), the paper's own translations are the control.

Methods

Data. Of the 517 Imvo issues, the 511 with at least 1,000 words per language are used. Chunks in the wrong language and text repeated across issues are left out, leaving 45,465 isiXhosa and 22,927 English chunks. Issues are split 357/77/77 (train/validation/test), stratified by year.

Spelling modernisation. A word list is built from 3.3M tokens of modern isiXhosa (mostly OPUS-100 (Zhang, Williams, et al., 2020); also Wikipedia and Vukuzenzele (Lastrucci, Rajab, et al., 2023)). *Aspiration* restores the *h* that 1880s spelling left out after *t, p, k, r, c, q, x*, but only when the old spelling is almost absent from the modern word list ($\leq$2 uses) and the new one is common in it ($\geq$5). *Prefix* joins prefixes written apart (*e Palamente*$\rightarrow$*ePalamente*) only when the joined form is in the list ($\geq$5 uses). The rules rewrite 9.1% of the 3.7M isiXhosa words, including 191 likely names (capitalised in $\geq$90% of uses). *Modern* applies both rules; *raw* is the original.

Embeddings and translated pairs. Each chunk is embedded with LaBSE and, for comparison, multilingual-E5 (Wang et al., 2024). Within each issue, an isiXhosa and an English chunk that are each other's closest match by cosine similarity form a pair if their similarity is $\geq$0.55 (311 test pairs; 10,289 test chunks). Of 100 sampled test pairs, a fluent isiXhosa speaker labelled 24 herself and reviewed 27 that a team member had labelled with AI translations and suggested labels. Of these 51, 31% were translations, 29% partial and 39% different, suggesting that the 0.55 threshold is too loose. Her own labels find more pairs different (54% vs 26%), so the reviewed ones may lean towards the AI, though Δ is similar (0.24, 0.26).

Removing the language signal (pair removal). Pair differences, which are mostly language, are stacked for training pairs and their top k directions are projected out of every embedding:

$$D = \left[x_i^{\mathrm{xh}} - x_i^{\mathrm{en}}\right]_{i \in P}, \qquad \tilde{x} = x - V_k^{\top} V_k x, \tag{1}$$

where V_k holds the top-k right singular vectors of D. A logistic regression, weighted so both sections count equally, is trained on training issues to predict each chunk's section; the difference between its BA on all test chunks and on test pairs gives Δ. Two settings are chosen on validation issues: the number of directions removed, k, is the smallest value in $\{0, 1, 2, 4, \ldots, 64\}$ that brings the floor to 0.55 or below (64 if none does, as here), and the regularisation strength

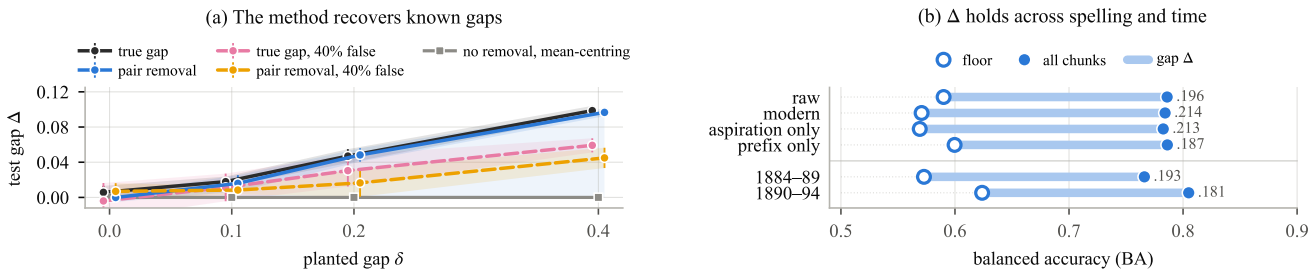


Figure 1: (a) Simulation: gap found vs gap planted (5 seeds, ±std.; dashed: 40% false pairs); pair removal is this method, mean-centring subtracts each language's mean. (b) Test issues, LaBSE: each line runs from the floor (open) to all-chunk BA (filled); its length is Δ.

$C \in \{10^{-2}, \ldots, 10^{3}\}$ is the one with the highest BA. **Spelling effects.** For H2, each spelling variant keeps the pairs and k chosen on the original text, so only the spelling changes. It is equivalent to raw if the 90% CI of its change in Δ lies within ±0.02. **Topic analysis.** Topics are k-means clusters of $\tilde{x}$, fitted without section labels. To test how much of the gap is subject matter alone, a *topic-only rule* assigns a chunk to the isiXhosa section if its topic has an isiXhosa share in training issues above the overall share (as in Figure S1, but refitted). This and the check of where errors fall were performed after the main results, so they are exploratory. **Synthetic check.** Each simulated chunk is a topic vector plus a language vector. A content difference is planted on purpose: simulated English chunks use all topics evenly, while a share δ of simulated isiXhosa chunks use only a quarter of them. Simulated pairs copy isiXhosa content into English, and a classifier on topic vectors alone measures the true difference. This checks the language removal and the floor, not LaBSE.

Results

The method finds planted gaps. In simulation (Figure 1a) the planted gap is recovered (0.097 vs true 0.099 at $\delta{=}0.4$); with 40% false pairs (recorded pairs that are not in fact translations) it shrinks (0.045 vs a true 0.059) rather than inflates.

The sections differ in content, not just language (H1). Of 311 test pairs, the classifier gives both halves one label for 201 (it cannot tell a text from its translation), correctly tells them apart for 83 and labels both the wrong way round for 27; the floor exceeds 0.5 only because 83 outnumber 27. Above this floor, Δ is 0.196 (0.167–0.224), supporting H1, and is positive in every Table S2 setting (smallest 0.046) and both periods (Figure 1b). On 16 hand-checked full translations the floor is 0.50, so loose pairs raise the floor and the main Δ errs low.

The BA gap is mostly subject matter. A topic-only rule reaches 0.661 on test chunks, but only 0.523 on translated pairs, whose two halves share a topic. Topic accounts for about 70% of the gap (0.138 of 0.196). The classifier also errs mostly on chunks about topics the other section usually covers: 34% of chunks, but 63% of errors. Topic is not everything, though: within one topic the classifier still reaches 0.73.

Spelling helps the tool read but does not detectably move Δ (H2). Aspiration cuts words missing from modern text from 31.5% to 24.5%, and translated pairs move closer (similarity +0.005). Its effect on Δ cannot be told apart from zero (+0.018, 90% CI −0.003 to +0.037; Figure 1b, Table S1; likewise for E5), and the floor's fall (0.019) is within noise. Every CI crosses ±0.02, so H2 is neither confirmed nor rejected.

Discussion

Subject matter drives the difference. The sections are known to differ (Schoots, Ngwendu, et al., 2023), so what is new is measuring how much once language is removed, and showing that cleaning the spelling does not detectably change the answer. The English section leans towards colonial policy: commissions, land laws and liquor licensing; the isiXhosa section towards community life: association meetings, sport, marriage, land sales and market prices (Figure S1). The classifier errs where one section printed what the other usually did: 23 of 24 English land-sale notices go to the isiXhosa section, and most isiXhosa land-law and liquor articles go to the English one. What remains within a topic (0.73) may be how each section catered for its readers.

Translation is an imperfect control. Most pairs land in the same section, yet the isiXhosa speaker noted the saying *isiXhosa asitolikwa*, "isiXhosa cannot be translated": even good translations adapt, turning "a feather in its cap" into the *indwe*, the crane feather of honour (Y. Siko, pers. comm.). This suggests that some meaning relies on cultural identity. If LaBSE misses this meaning too, the classifier cannot use it to tell more pairs apart, so the floor controls content only roughly.

Modernisation and the historical record. Modernising is not needed for this conclusion, yet its rules rewrite how people named themselves in 1884: a people (*Xosa*), a community (*Tembu*), a town (*Mtata*) and God (*Tixo*). The tools read English more easily too (2.11 vs 1.26 LaBSE tokens per word), which may tempt researchers to study only the English half (Nekoto, Marivate, et al., 2020). **Limitations.** Only 51 pairs were hand-checked and no one checked the rewrites. Topic checks use LaBSE only, and the method measures what was printed but cannot say why. Future archive studies should report the floor under both spellings.

Contribution Statements

We, Gopolang Mohlakola, Tawananyasha Kadango and Lily de Melo, worked together fairly on this project.

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    Justification: The code, normaliser and labelling guide are documented in README.txt and labels/LABELLING_GUIDE.md.

    Guidelines:

    - The answer NA means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
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14. **Crowdsourcing and Research with Human Subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [NA]

    Justification: No crowdsourcing or research with human subjects. A fluent isiXhosa speaker labelled 24 sampled translated pairs herself and reviewed 27 more that a team member had labelled with AI translations and suggested labels (translation, partial or different), following labels/LABELLING_GUIDE.md; no personal data was collected, and the instructions are released with the labels.

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    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [NA]

    Justification: No human subjects: the texts are newspapers from 1884–1894, and the archive's authors note that no ethics review was needed because the materials discuss no living subjects (Schoots et al., 2023).

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Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☐
2. Sourcing Related Work (summarising, identifying sources) ☐
3. Methods and Experiment Design (experiment setup, model tuning) ☐
4. Data Analysis (presentation, coding, interpretation) ☐
5. Theoretical Development (theorem proving, conceptual analysis) ☐
6. Code Development (generating algorithms, writing scripts) ☐
7. Presentation (rendering graphics, formatting) ☐
8. Editing (grammar, readability) ☐
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If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number:

Candidate signature:

Date:

Supplementary material

Two tables and a figure that support claims in the body, from the saved results in `results/`.

Table S1: Section classifier on the test set, LaBSE (balanced accuracy). Spelling variants barely move Δ. The last column is each variant's change in Δ from the original raw text, with its 90% CI in brackets: the true change likely lies between the lower and upper ends (1,000 resamples of test issues).

Variant	All	Floor	Δ	$\Delta - \Delta_{\text{raw}}$ (90% CI: low, high)
raw	.786	.590	.196	–
modern	.784	.571	.214	+.018 (−.003, +.037)
aspiration only	.783	.569	.213	+.017 (−.001, +.038)
prefix only	.786	.600	.187	−.009 (−.024, +.006)

Table S2: The gap holds across checks. All rows use the test set, LaBSE and original spelling unless otherwise stated. The brackets give the 95% CI, where available: the true gap likely lies between the lower and upper ends (resampled over test issues). kNN is a cosine k-nearest-neighbour classifier, which fits no weights. The topic-only rule uses topics alone, refitted from Figure S1 (using the same code and seed; cluster sizes differ slightly).

Setting	All	Floor	Δ (95% CI: low, high)
Main result (k=64)	.786	.590	.196 (.167, .224)
E5 encoder	.877	.618	.258 (.216, .304)
kNN classifier	.851	.703	.149
kNN classifier, E5	.599	.553	.046
Topic-only rule	.661	.523	.138
Hand-checked full translations (16 pairs)	.786	.500	.286 (.142, .412)

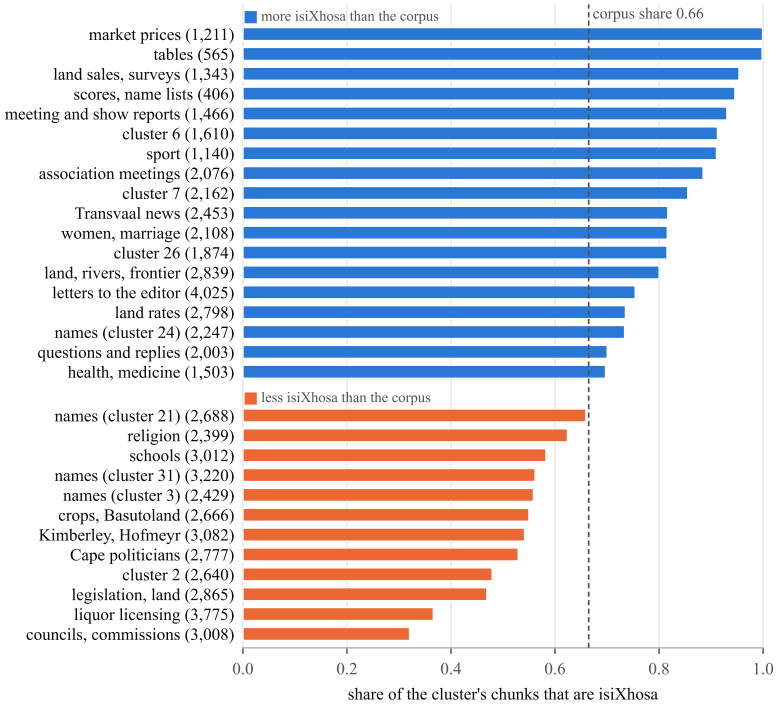


Figure S1: What each section usually wrote about. The share of each topic's chunks that are isiXhosa, for 30 topics, each with its chunk count. Blue topics are more isiXhosa than the overall share (0.66), while orange ones are less. The topics are k-means clusters of the language-removed embeddings, named by the authors from each cluster's most distinctive words (those most over-represented in it compared with the rest). Clusters labelled *names* hold mostly personal and place names, and numbered ones have no clear subject. The topics overlap (silhouette score 0.008, where 0 means fully overlapping), so they describe content rather than measure it.